Speed, Distance and Time — Paper B

MEDIUM · 40 MINUTES · 30 MARKS

NAME DATE SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a unit. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- Reminder: speed = distance ÷ time, and to convert m/s to km/h multiply by 3.6.

SECTION 1 — READING A JOURNEY

1. A runner's journey is recorded below.

Time (s):	0	20	40	60	80
Distance (m):	0	40	80	80	30

(a) Calculate the speed during the first 40 seconds. [2]

(b) Describe what happens between 40 s and 60 s, and state how you know. [1]

(c) Describe what happens between 60 s and 80 s, and calculate the speed during it. [2]

2. Sketch a distance–time graph for this journey, labelling both axes with units: A car travels 300 m in 30 s at a steady speed, stops for 20 s, then travels a further 100 m in 20 s. [4]

3. Using the journey in question 2, describe it in three sentences. Every sentence must contain a number. [4]

SECTION 2 — AVERAGE SPEED

4. A car travels 20 km in 30 minutes, then a further 30 km in 1 hour. Calculate the average speed for the whole journey in km/h. [4]

5. A cyclist rides 15 km in 30 minutes, rests for 15 minutes, then rides 10 km in 15 minutes. Calculate the average speed for the whole journey in km/h. [4]

6. Explain why stopping for a rest lowers the average speed of a journey, even though the cyclist does not travel any slower when moving. [3]

SECTION 3 — COMPARING

7. Which is faster: 12 m/s, or 40 km/h? Show your working and state which one. [3]

8. Explain what the gradient of a distance–time graph represents, and describe how you would calculate it from a graph. [3]

End of paper. Check every answer has a unit or a form the question actually asked for.